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ORIGINAL PAPER

Performance evaluation and bias correction of gridded precipitation products over Arun River Basin in Nepal for hydrological applications

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Abstract

An accurate estimation of precipitation amount is crucial for various studies and planning related to water resource management, effective flood prediction and warning systems, agriculture, climatic research, and disaster risk management. However, due to the sparse and uneven distribution of ground-based precipitation gauges over rugged terrain, accurate and consistent measurement is inadequate in many developing and mountainous countries like Nepal. Therefore, satellite-based precipitation products (SPPs) and interpolation-based gridded data are considered as a vital source of precipitation estimation, which may serve as crucial inputs for a wide range of hydrological applications. However, in the absence of quality assessment, applications of these products pose uncertainty. This study evaluated the performance of three SPPs, i.e., CHIRPS V2.0, PERSIANN CDR, and MSWEP V2.8, and a ground-based gridded precipitation product APHRODITE on daily, monthly, and annual scales at ten rain gauges over the Arun River Basin. The performance of the precipitation products was evaluated from 1983 to 2014 using several statistical categorical and continuous indices. Our results show APHRODITE and MSWEP V2.8 are comparatively better than CHIRPS V2.0 and PERSIANN CDR in the study area. We finally applied the bias correction of the selected products using a linear scaling method, where daily precipitation data were corrected using a monthly correction factor. We find all SPPs have improved after the bias correction. The method is scalable and applicable in other river basins across the country and beyond Nepal.

1 Introduction

Precipitation is a significant component of the hydrological cycle that can impact the Earth's meteorological system if its patterns are changed (Amorim et al. 2020; Ji et al. 2020; Venkatesh et al. 2020). The energy-water cycle and basin hydrology fundamentally depend upon precipitation patterns, amount, and intensity (Talchabhadel et al. 2021). Therefore, a sound understanding of precipitation variability is crucial in water resource management, effective flood prediction and warning systems, agriculture, climatic research, and disaster risk management studies (Bajracharya et al. 2015). However, the density of precipitation gauges for recording precipitation is still coarse across the globe (Kidd et al. 2017) in general, in low-economy countries in particular. Ground-based precipitation stations provide highly accurate results; however, sparse and uneven distribution of precipitation gauges over unpopulated regions and rugged terrain makes it difficult to record accurate data for climate and hydrological applications (Venkatesh et al. 2020). The World Meteorological Organization (WMO 2008) recommended density of ground-based gauge stations in hilly/undulating areas is 1 gauge per 575 km^2 and 250 km^2 for mountainous areas, respectively. Some of the reasons for the relatively lower density of precipitation gauging stations for countries, like Nepal, are wide topographic variations, technological constraints, and economy (Talchabhadel et al. 2021). Therefore, SPPs are increasingly becoming an important source of rainfall data to overcome the aforementioned limitations (Bui et al., 2019; Demb´el´e et al., 2020, as cited in Talchabhadel et al. 2021).

Numerous SPPs are available with high spatial resolutions, along with different spatial and temporal coverage. They can be divided into four categories (Chen et al. 2020; Mazzoleni et al. 2019): ground-based rain gauge datasets, raw satellite datasets, satellite reanalysis gauge datasets,

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and satellite gauge-adjusted datasets. Among various SPPs, the following four SPPs are selected for investigating their behavior: Asian Precipitation-Highly Resolved Observational Data Integration Towards Evaluation of Water Resources (APHRODITE) (Yatagai et al. 2012), Climate Hazards Group InfraRed Precipitation with Stations version 2.0 (CHIRPS V2.0) (Funk et al. 2015), Multi-Source Weighted-Ensemble Precipitation version 2.8 (MSWEP V2.8) (Beck et al. 2019), and Precipitation Estimation from Remotely Sensed Information Using Artificial Neural Networks-Climate Data Record (PERSIANN CDR) (Miao et al. 2015). These products are selected based on their long-term data series.

Several metrics are developed to evaluate their accuracy and suitability for climatic characterization and input for hydrological modeling (Venkatesh et al. 2020). Two widely used approaches for validation/evaluation of SPPs are the following (Tong et al. 2014): (a) direct comparison of SPPs with observed gauge data (Abd Elhamid et al. 2020; Ghaemi et al. 2021; Hamal et al. 2020; Masood et al. 2020; Sharma et al. 2020a; Zhang et al. 2020) and (b) evaluation through hydrological modeling scheme based on their anticipating ability of streamflow (Amorim et al. 2020; Belayneh et al. 2020; Ji et al. 2020; Talchabhadel et al. 2021; Wang et al. 2021).

SPPs are likely to be associated with various uncertainties (Dinku et al. 2010; Tian et al. 2010), such as the errors during the retrieval process and bias correction processes (Sun et al. 2016). Hence, these errors should be reduced before using the product for the specific application (Belayneh et al. 2020). For bias correction of the SPPs, Chen et al. (2013) classified bias correction methods into two categories: (a) mean-based approaches and (b) distribution-based approaches. Müller & Thompson (2013) implemented bias correction of Tropical Rainfall Measuring Mission (TRMM) satellite product by stochastic modeling approach over Nepal. Moazami et al. (2014) analyzed the uncertainty and adjusted the bias of two SPPs (PERSIANN and TMPA-3B42) by developing a copula-based ensemble simulation method. Talchabhadel et al. (2021) used the linear scaling method of bias correction to correct biases of SPPs where monthly correction factors (CF = gauge/SPPs) were calculated by comparing the mean monthly precipitation of the ground-based gauge and SPPs. Among these studies, the most simple and scalable method is the linear scaling method which belongs to the mean-based approach (Ji et al. 2020).

Only limited studies focused on evaluating SPPs and rain gauge data in Nepal. These studies are performed over Nepal (Athey et al. 2015; Duncan & Biggs 2012; Hamal et al. 2020; Islam et al. 2010; Krakauer et al. 2013; Müller & Thompson 2013; Nepal et al. 2021; Sharma et al. 2020a, b) or on large catchment rivers like the Koshi River (Hossain et al. 2016; Shrestha et al. 2017b), Narayani River (Ghaju & Alfredsen 2016; Shrestha et al. 2011), Karnali River (Khatakho et al. 2021), or on medium-catchment rivers, like West Rapti (Talchabhadel et al. 2021) and Bagmati (Bajracharya et al. 2017; Shrestha et al. 2008). Also, most of these studies have focused on either analyzing a single SPP's performance or evaluating the efficiency of a few SPPs in streamflow simulation with a smaller time frame. For example, Shrestha et al. (2017b) performed a study over the Koshi River, which evaluated only two satellite products of Climate Hazard Group (CHG) at a monthly scale. Different SPPs with long-term data series should also be evaluated on daily, monthly, and annual time scales to evaluate their performance.

The Arun River Basin (ARB), a tributary of the Koshi River Basin that feeds to the Ganges, hosts several hydropower projects either under construction or planned. Therefore, reasonable estimation of precipitation and hydrology is crucial for those water infrastructure projects. However, precipitation observation within the ARB is sparse. Furthermore, a large area of the ARB is within China, with limited access to those valuable data. In this context, exploring the possibility of using suitable SPP is vital. However, there are no such studies performed so far. Therefore, this study aimed to evaluate the performance of a few SPPs with longer datasets (1983–2014) with respect to the ground-based rain gauge datasets and correct them using a scalable method. Therefore, the specific objectives of this study are as follows: (i) evaluation of raw SPPs at different temporal scales (daily, monthly, and annual) and (ii) evaluation of SPPs after bias correction for various applications. We present the data, methods, and characteristics of the study area in "Materials and methods." "Results and discussion" provides the results of this study and discussion on their implications. Finally, the last section offers a synopsis and conclusions.

2 Materials and methods

2.1 Study area

The Arun River originates in the Tibet autonomous region of China with the greatest snow and ice-covered area than any other Nepalese river basin (Kattelmann 1990). In the Nepal Himalaya, the Arun River is the longest river with a total length of 510 km, of which only 155 km lies within Nepalese territory (Shrestha 1989). Geographically, the ARB lies in eastern Nepal, with its coordinates extending from about 27 to 29° North latitudes and 85.5 to 89° East longitudes. The total catchment area of the Arun River at its confluence with the Koshi River is about 33,523 km^2, among which about 28,368 km^2 lies in the Chinese territory and about 5155 km^2 lies within the Nepalese territory (Fig. 1). About 84% of the ARB's catchment

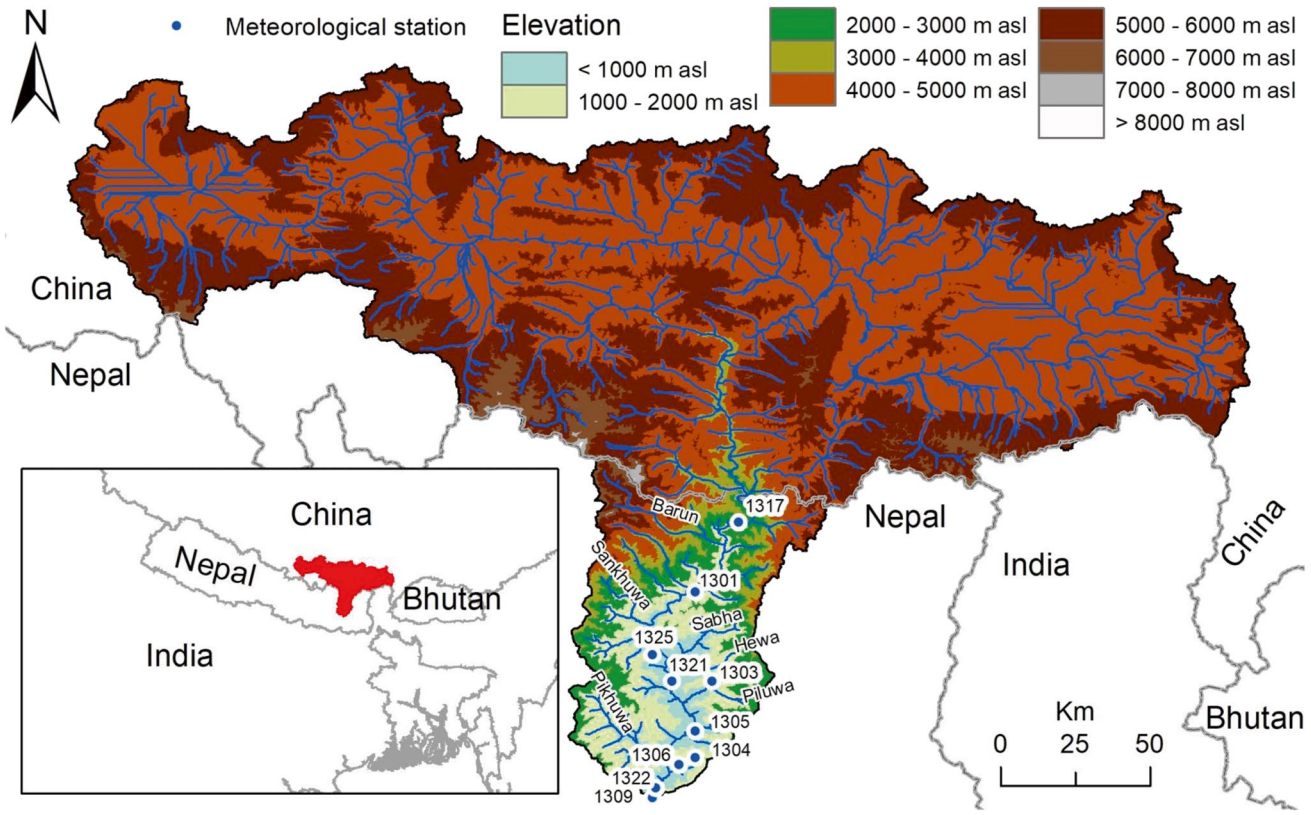


Fig. 1 Location, meteorological stations, and elevation details of Arun River Basin

area lies in the rain shadow of the Himalayas, resulting in less discharge contribution to the Arun river system (Kattelmann 1990). More than 70% of the Arun's total discharge is contributed by the catchment of about 5155 km^2 (about 16% of the total catchment of ARB) inside Nepal (JICA 1985).

The Advanced Land Observing Satellite (ALOS), global Digital Surface Model (DSM) version 3.2 with a horizontal resolution of approximately 30 m, has been used as the topographic dataset in this study. The variation of elevation in the ARB ranges from 117 to 8768 masl. In the low-lying valleys, the climate is tropical. As the altitude rises towards the crest of the Himalayas, the region gradually passes into the Temperate, the Alpine, and the Arctic climate (Karki et al. 2015). About 60–80% of annual rainfall, as recorded by the weather stations maintained by the Department of Hydrology and Meteorology (DHM), Nepal, in the ARB, occurs during the peak Indian Summer Monsoon (ISM) months (June–September) (Bookhagen & Burbank 2010; Olen et al. 2014). The major tributaries of Arun River are Hewa Khola, Piluwa Khola, Pikhuwa Khola, Barun River, Sabha Khola, and Sankhuwa Khola among other minor tributaries.

2.2 Datasets description

In this study, we have selected three SPPs (CHIRPS V2.0, MSWEP V2.8, and PERSIANN CDR) and one observation-based gridded (APHRODITE) dataset and ten ground-based gauges over a 32-year period from 1983 to 2014. During the chosen period, both gridded datasets and ground-based gauge datasets are available without large data gaps. We chose the stations with a data gap of less than 10% during the study period.

2.2.1 Ground-based rain gauges

The historical records of daily precipitation datasets for 1983–2014 were gathered from ten ground-based meteorological stations within ARB. The DHM, Government of Nepal, maintained these meteorological stations. Details on the ground-based rainfall observations are available at Talchabhadel et al. (2017). The distribution of these meteorological stations over the ARB is shown in Fig. 1, and their basic information is summarized in Table S1 in the supplement section. Data quality assessment has been performed for these stations, and rain gauges with less than

10% of missing data in the daily precipitation time series from available data have been selected (Amorim et al. 2020). The detail of meteorological stations with the percentage of missing data is shown in Table 1.

Furthermore, we employed the normal rainfall (NR) method to interpolate the missing rain gauge data series. The data length of 1983–2014 was selected based on the availability of continuous data series from 1983 onwards. The data length was truncated in 2014 because some ground-based rain gauges have a whole year of missing data in 2015 and 2016.

2.2.2 Satellite precipitation products

Besides ground-based rain gauge, there are two other alternate sources of precipitation datasets, i.e., weather radar and satellite (Talchabhadel et al. 2021). Due to the unavailability of weather radar, three high-resolution SPPs (CHIRPS V2.0, MSWEP V2.8, and PERSIANN CDR) and one observation-based gridded (APHRODITE) were evaluated in this paper. Basic information about the spatial resolution, coverage, and available periods for all four SPPs is shown in Table 2.

APHRODITE The APHRODITE datasets were developed using ground-based precipitation gauge data that has been collected from the national hydrological and meteorological departments of various Asian countries (Yatagai et al. 2012). The APHRODITE products are available in high resolution (0.25° and 0.5°), yield daily gridded precipitation data all over the Asian region, and can be downloaded freely from http://aphrodite.st.hirosaki-u.ac.jp/download/ (Ji et al. 2020; Tan et al. 2015). Previously, over the monsoon Asia region, APHRODITE-V1101 was available from 1951 to 2007, and the later APHRODITE-V1101EX_R1 was released, extending the time range from 2007 to 2015 (Ji et al. 2020). For this study, we have combined two versions with a resolution of 0.25°, and the subset data from 1983 to 2014 was used for evaluation.

CHIRPS V2.0 CHIRPS V2.0 is a quasi-global (50° S–50° N) and high-resolution (0.05°) rainfall dataset developed by Climate Hazards Group (CGH) (Funk et al. 2015) available from 1981 to the present days (Talchabhadel et al. 2021) with various temporal series of rainfall such as daily, pentad, and monthly (Wang et al. 2021). For the construction of the CHIRPS V2.0 dataset, various data sources were blended together like monthly Climate Hazards Group Precipitation Climatology (CHPclim), satellite-based infrared observations from several sources, and long-term in situ gauge observations. Funk et al. (2015) provide detailed information on the algorithm used to develop CHIRPS V2.0 datasets. The CHIRPS datasets are available freely at https://data.chc.ucsb.edu/products/CHIRPS-2.0/. In this study, CHIRPS V2.0 datasets of daily temporal series with 0.05° spatial resolution were evaluated.

MSWEP V2.8 MSWEP V2.8 is developed by combining the gauge-based, satellite-based, and reanalysis-based data to generate decent global precipitation datasets (Beck et al. 2019; Sharifi et al. 2019). The MSWEP V2.8 dataset is available from 1979 to the present day. The dataset is available at the spatial resolution of 0.1° and temporal resolutions of three hourly, daily, and monthly (Beck et al. 2019; Sharifi et al. 2019). Evaluation of the newest version of this product (V2.8) is done in this study. The data license to download the MSWEP dataset can be obtained freely via http://www.gloh2o.org/mswep/.

PERSIANN CDR PERSIANN CDR is one of the new bias-adjusted SPP made available to the public and can be acquired from 1 January 1983 to today (Ashouri et al. 2015). At present, PERSIANN CDR is available at a high spatial resolution of 0.25°, and daily temporal resolution for the latitude band from 60° S–60° N (Jiang et al. 2017); the data can be downloaded directly from https://chrsdata.eng.uci.edu/. It is estimated using the PERSIANN algorithm on Gridded Satellite (GridSat-B1) infrared data as the primary input satellite data (Tan & Santo 2018). It uses an ANN model to convert rainfall rates using infrared brightness temperature data developed by geostationary satellites. PMW observations are used from low-orbital satellites to update their parameters (Hsu et al. 1997; Tan et al. 2015). Its reliability is improved by calibrating the estimated precipitation data using monthly GPCP version 2.2 products (Ashouri et al. 2015). The detailed description of the PERSIANN family is discussed in Nguyen et al. (2018). In this paper, data on a daily scale from 1983 to 2014 are used for evaluation.

2.3 Evaluation of SPPs

In this study, we performed the comparison and evaluation of three SPPs and a ground-based gridded precipitation product versus the ground-based observed datasets maintained by DHM, Nepal. For comparison of SPPs and ground-based observed datasets, three approaches (pixel to pixel, point to pixel, and point to point) have been widely practiced. In the study performed by Shrestha et al. (2011), the spatial interpolation method Kriging was used to estimate spatial rainfall from rain gauge networks and then carried out pixel-to-pixel comparisons (Shrestha et al. 2017b). Other researchers (e.g., Shrestha et al. (2017b), Ghaju and Alfredsen (2016), and Mashingia et al. (2014)) extracted an areal average of rainfall estimates over the pixel in which the observation gauge is located and then carried out point-to-pixel comparisons. This study compared the gauge observed value with the SPP's grid value nearest to the ground-based

Table 1 List of meteorological stations with missing data

S. no	Station number	Station no	Available data	Total number of days	Number of missing days	Percentage of missing days	No. of missing days in rainy season	Percentage of missing days in rainy season	Missing data
1	Num	1301	1979–2019	14,975	1180	7.9	274	1.8	(Nov-Dec) 1999; (Jan-Feb/Dec) 2001; (Apr/Nov-Dec) 2002; (Dec) 2003; (Nov-Dec) 2007; (Apr) 2010; (Sep-Dec) 2011; (May) 2013; (Oct-Dec) 2014; 2016–2017
2	Chainpur (east)	1303	1948–2019	26,298	68	0.3	31	0.1	(Jan) 1971; (Jul) 1975; (May 1–6) 2019
3	Pakhribas	1304	1976–2019	16,071	397	2.5	213	1.3	(Jun-Nov) 1989; (Sep-Oct) 1997; (Aug-Dec) 1998
4	Leguwa ghat	1305	1979–2019	14,975	334	2.2	122	0.8	(Jan/Oct-Dec) 2001; (May-Dec) 2016
5	Munga	1306	1948–2019	26,298	1538	5.9	535	2.0	(May-Jun/Oct-Nov) 1948; (Feb-May) 1949; (Jul-Nov) 1950; (May-Aug) 1951; (Dec) 1955; (Jul-Dec) 1958; (Jan) 1971 (Oct) 2013; 2015–2016; (Sep) 2017; (Jan-Mar) 2018
6	Tribeni	1309	1949–2019	25,932	312	1.2	30	0.1	(Dec) 1950; (Apr-May/Sep-Nov) 1951; (Apr) 1952; (Jan/Dec) 2001; (Jan-Feb) 2009; (Jan) 2010; (Apr-May) 2012; (Apr-May) 2014
7	Chepuwa	1317	1979–2019	14,975	339	2.3	166	1.1	(Dec) 2001; (May-Jul) 2004; (Mar-Apr) 2007; (Nov-Dec) 2009; (Jan) 2010; (Sep-Oct) 2011; (Apr-Sep) 2012
8	Tumlingtar	1321	1977–2019	15,705	78	0.5	0	0	(Dec) 2001; (Jan) 2010; (Mar-Apr) 2016
9	Machuwaghat	1322	1949–2019	25,932	454	1.8	36	0.1	(Jan-Feb/Apr/Nov) 1951; (Jan-Jun)1977; (Jan/Dec) 2001; (Sep-Oct) 2004; (Feb-Mar) 2011; (Dec) 2014; (Jan-Feb) 2016
10	Dingla	1325	1979–2019	14,975	118	0.8	14	0.1	(Dec) 2000; (Jan/Dec) 2001; (Jan-Feb/Sep-Oct) 2008

Table 2 Overview of four gridded precipitation datasets evaluated in this study

S. no	Precipitation datasets	Type	Spatial/ temporal resolution	Spatial coverage	Temporal coverage	Data reference	Website
1	APHRODITE	Gauge-based	0.25°/daily	Monsoon Asia 60–150° E and 15° S–55° N	1951–2015	Yatagai et al. 2012	http://aphrodite.st.hirosaki-u.ac.jp/download/
2	CHIRPS V2.0	Reanalysis product	0.05°/daily	50° S–50° N	1981-NRT	Funk et al. 2015	https://data.chc.ucsb.edu/products/CHIRPS-2.0/
3	MSWEP V2.8	Reanalysis product	0.1°/daily	Global	1979-NRT	Beck et al. 2019	http://www.gloh2o.org/mswep/
4	PERSIANN CDR	Gauge adjusted	0.25°/daily	60° S–60° N	1983-NRT	Miao et al. 2015	https://chrsdata.eng.uci.edu/

rain gauge. The nearest grid value was established by calculating the minimum of the square of the distance between the gauge and the grid point. Then point-to-point comparison was made. Evaluation of the SPPs versus observed gauge data was done using five categorical statistical indices such as accuracy (ACC), probability of detection (POD), false alarm ratio (FAR), critical success index (CSI), and frequency bias index (FBI) along with five continuous statistical indices such as root-mean-square error (RMSE), coefficient of determination (R^2), mean error (ME), mean absolute error (MAE), and relative bias (RB). These indices are further elaborated in the following subsections.

2.3.1 Categorical statistical indices

Equations 1 to 5 represent five widely applied categorical statistical approaches such as ACC, POD, FAR, CSI, and FBI, which have been used to evaluate the precipitation detection capability of each SPP. The ACC represents the closeness of the estimated values given SPPs to the actual value measured by ground-based rain gauge (Tan et al. 2015). The POD is the ability of the SPPs to recognize the real occurrences of rainfall encountered by the ground-based rain gauge; the FAR is the indicator that measures the rainfall observed by the SPPs even when a ground-based rain gauge measured no rainfall, and the CSI measures the portion of a ground-based rain gauge's event that was correctly estimated by the SPPs (Mashingia et al. 2014; Talchabhadel et al. 2021; Tan et al. 2015; Tan & Santo 2018). The value for ACC, POD, FAR, and CSI ranges from 0 to 1. The perfect score for ACC, POD, and CSI is 1, whereas the perfect score for FAR is 0. Finally, the FBI represents a simple bias between precipitation events estimated by SPPs and the precipitation events measured by ground-based rain gauge (Talchabhadel et al. 2021). The value of FBI ranges from 0 to ∞; if FBI values > 1, it indicates an overestimation of precipitation event detection, and if < 1, it indicates an underestimation of precipitation event detection (Talchabhadel et al. 2021).

$$ACC = \frac{A + D}{n} \tag{1}$$

$$POD = \frac{A}{A + C} \tag{2}$$

$$FAR = \frac{B}{A + B} \tag{3}$$

$$CSI = \frac{A}{A + B + C} \tag{4}$$

$$FBI = \frac{A + B}{A + C} \tag{5}$$

where A is the observed rainfall that has been detected correctly by the SPPs, B is the observed rainfall detected incorrectly by the SPPs, C is the observed rainfall that has been missed by the SPPs even when the gauge has shown rainfall event, D represents a correct estimation of no rain by SPPs, and n is the total number of events.

2.3.2 Continuous statistical indices

Equations 6 to 10 show five different continuous statistical approaches, RMSE, R^2, ME, MAE, and RB, which were used to evaluate and compare the performance of SPPs with respect to the ground-based observed data. First, the average magnitude of the deviation of SPPs from the ground-based observed data is measured as the RMSE; a smaller RMSE value indicates SPP estimates are close enough with the observed data (Hamal et al. 2020; Tan & Santo 2018). Second, the grade of linear correlation between the SPPs and the ground-based observed data is evaluated using R^2, whose value ranges from − 1 to 1. Zero

indicates no correlation between the SPPs and ground-based observed data, whereas 1 and − 1 indicate perfect positive and negative correlations, respectively (Tan & Santo 2018). Third, the ME and MAE measure the average magnitude of the error between the SPPs and the ground-based observation. The major difference between the ME and MAE is that a ME can give positive and negative results, whereas MAE gives positive results only. Zero value indicates that the data of SPPs and ground-based observation are a perfect match. A higher value of ME and MAE indicates a higher error in the estimated and observed data. Finally, the systematic bias of the SPPs with the observed data is evaluated using RB, which can either be positive or negative. If the RB value is positive, it indicates an overestimation of the rainfall amount, whereas if the value is negative, it indicates an underestimation of the rainfall amount (Tan & Santo 2018).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(S_i - G_i)^2}{N}} \tag{6}$$

$$R^2 = \frac{\sum_{i=1}^{n}\left(G_i - \overline{G}\right)\left(S_i - \overline{S}\right)}{\sqrt{\sum_{i=1}^{n}\left(G_i - \overline{G}\right)^2}\sqrt{\sum_{i=1}^{n}\left(S_i - \overline{S}\right)^2}} \tag{7}$$

$$ME = \frac{\sum_{i=1}^{n}(S_i - G_i)}{N} \tag{8}$$

$$MAE = \frac{\sum_{i=1}^{n}|S_i - G_i|}{N} \tag{9}$$

$$RB = \frac{\sum_{i=1}^{n}(S_i - G_i)}{\sum_{i=1}^{n}G_i} \times 100 \tag{10}$$

where G_i and $\overline{G}$ are the individual and averaged observed precipitation provided by ground-based data, respectively. S_i and $\overline{S}$ are the individual and averaged estimated precipitation provided by satellite-derived products, respectively. And N is the total number of measurements. The observed and estimated precipitation is considered fully consistent without the uncertainty of satellite rainfall products if the values of ME, MAE, RMSE, and $RB = 0$ and $R^2 = 1$. Generally, SPPs are considered a reliable precipitation measurement source with an R^2 value higher than 0.7 and an RB value ranging between − 10% and + 10% (Brown, 2006; Condom et al., 2011 as cited in Tan & Santo 2018).

Furthermore, Eqs. 11 and 12 show the standard deviation (SD) and centered root-mean-square error (CRMSE). These values, along with the correlation coefficient value, were used to plot the Taylor diagram, which visually summarizes the closeness between the observation datasets and the datasets of the SPPs (Taylor 2001).

$$SD = \sqrt{\frac{1}{N}\sum_{i=1}^{n}\left(S_i - \overline{S}\right)^2} \tag{11}$$

$$CRMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{n}\left[\left(S_i - \overline{S}\right) - \left(G_i - \overline{G}\right)\right]^2} \tag{12}$$

where G_i and $\overline{G}$ are the individual and mean of observed rainfall data, respectively. S_i and $\overline{S}$ are the individual and mean of rainfall data provided by SPPs, respectively. And N is the total number of datasets.

2.4 Correcting biases in the selected SSPs

Bias correction can be described as the process of scaling satellite precipitation estimates with reference to the ground-based rain gauge data so that the systematic errors in the satellite precipitation estimates can be eliminated for its further processing (Belayneh et al. 2020). Even though linear scaling is the simplest method, it is equally effective for assessing resources as more complex bias correction methods (Shrestha et al. 2017a). We chose this method as it is easy to implement and upscale to other regions (Ji et al. 2020). First, monthly correction factors (CF = gauge/SPPs) were calculated as the ratio between the observed precipitation and the SPPs estimate on the mean monthly scale (Talchabhadel et al. 2021) for individual stations present in ARB. Every station's raw daily precipitation data is then multiplied with the calculated monthly correction factors (Talchabhadel et al. 2021). The correction factors for individual stations have been calculated. However, if the data quality of the ground-based rain gauge is poor, there may be an unreal representation of SPPs estimates (Talchabhadel et al. 2021). Therefore, care should be taken while performing the data quality assessment of the gauge data.

3 Results and discussion

3.1 Performance of selected raw SPPs

3.1.1 Performance for daily precipitation

We evaluated the performance of the SPPs at a daily scale for individual ground-based stations available in ARB. The categorical statistical indices of daily precipitation comparing the precipitation gauges and the SPPs from 1983 to 2014 are shown in Table S2 in the supplement section. Figure 2 displays the boxplot of the categorical statistical indices of daily precipitation events in ARB (1983 to 2014).

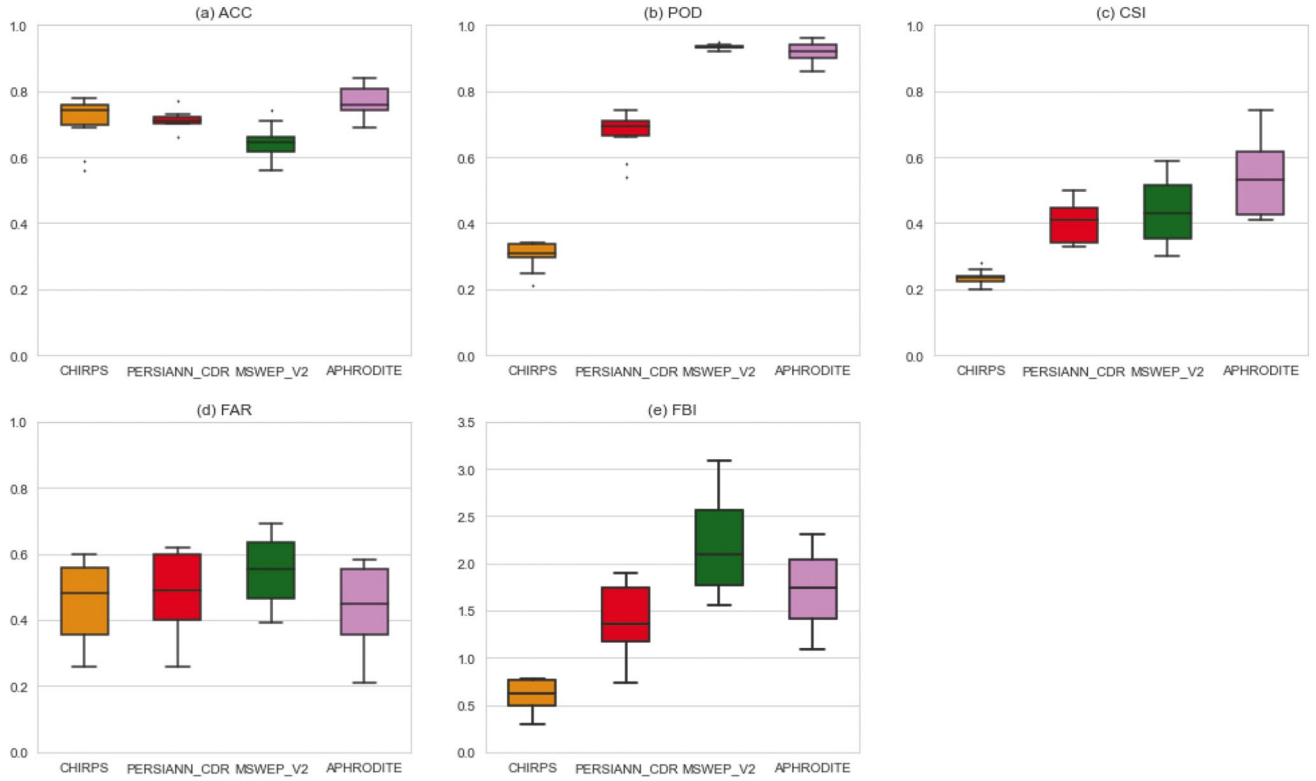


Fig. 2 Categorical statistical indices of daily precipitation events in ARB (1983 to 2014)

APHRODITE and PERSIANN CDR generated higher ACC values than CHIRPS V2.0 and MSWEP V2.8. However, CHIRPS V2.0 performed well in the rain gauge station, located below 500 masl (ACC values > 0.74 for Leguwa ghat, Tribeni, Tumlingtar, and Machuwaghat, as shown in Table S2). APHRODITE and MSWEP V2.8 performed better than CHIRPS V2.0 and PERSIANN CDR in the case of POD. The performance of CHIRPS V2.0 is much lower in all stations, as shown by POD. PERSIANN CDR performed poorly in stations Chepuwa and Num, whose elevations are 2590 and 1497 masl, respectively. All of the SPPs reported significant false alarms. All of the SPPs generated high FAR in stations whose elevations are less than 2000 masl. In the case of FAR, APHRODITE showed slightly good results than the other SPPs. CHIRPS V2.0 showed lower CSI values in all stations, and APHRODITE revealed higher CSI values in two stations, Num (1497 masl) and Chepuwa (2590 masl). APHRODITE performed better compared to other SPPs in terms of CSI also. FBI values for CHIPRS V2.0 are less than 1 for all stations, indicating underestimation of precipitation detection events. As shown by PERSIANN CDR, FBI value is less than 1 for two stations, Num (1497 masl) and Chepuwa (2590 masl), and greater than 1 for the remaining eight stations, indicating the mixed pattern of under- and overestimation of the events. Similarly, for MSWEP V2.8 and APHRODITE, FBI values are greater than 1 for all stations, indicating overestimated precipitation detection events.

Performance evaluation based on continuous statistical indices of various SPPs at a daily scale compared to ground-based stations present in ARB from 1983 to 2014 was tabulated in Table S3 in the supplement section. Accordingly, Fig. 3 displays the boxplot of continuous statistical indices of daily precipitation events in ARB (1983 to 2014).

The RMSE values for all four SPPs ranged from a minimum of 6.53 to a maximum of 28.91 mm/day, indicating that all four SPPs have daily biases, either positive or negative, in all stations. APHRODITE performed better with low RMSE values than other SPPs, whereas CHIRPS V2.0 performed poorly with high RMSE values. APHRODITE shows a higher correlation with the observed data with R^2 values ranging from 0.52 (for station Chepuwa at an elevation of 2590 masl) to 0.71 (for station Pakhribas at elevation 1680 masl), which is better compared to CHIRPS V2.0 (0.17 to 0.28), PERSIANN CDR (0.2 to 0.31), and MSWEP V2.8 (0.32 to 0.48). For all four SPPs, the ME value ranges from − 9.43 to 2.09, indicating lower error between estimated and observed data. Similarly, the presence of lower error in all four SPPs is indicated by the MAE value ranging from 2.9 to 13.79. Moreover, the RB of all SPPs has a range of negative to positive values indicating the mixed pattern of

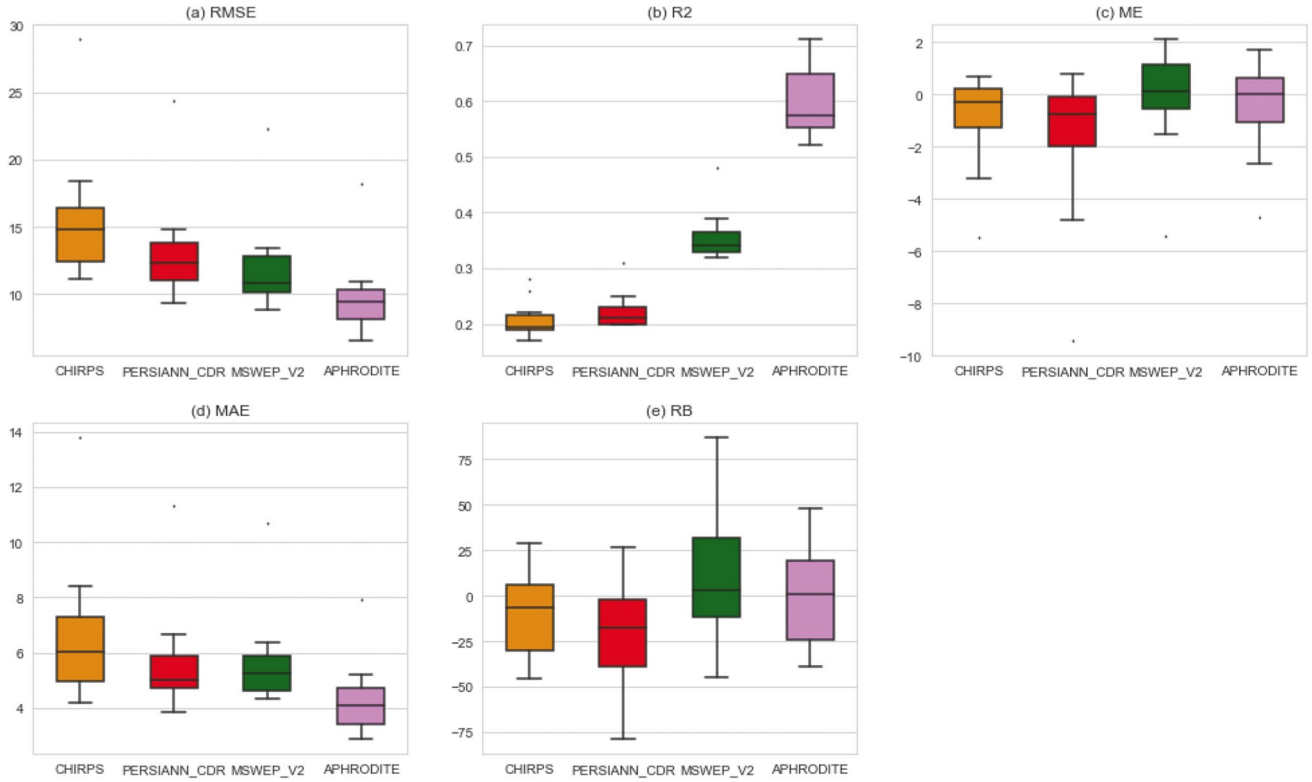


Fig. 3 Continuous statistical indices of daily precipitation events in ARB (1983 to 2014)

under- and overestimation of the rainfall event. All SPPs have shown underestimation of the rainfall event in Num (1497 masl) and Chepuwa (2590 masl). Performance evaluation using categorical and continuous statistical indices on a daily scale gives us a clear picture of APHRODITE performing better than CHIRPS V2.0, PERSIANN CDR, and MSWEP V2.8 in all stations except for stations Num and Chepuwa.

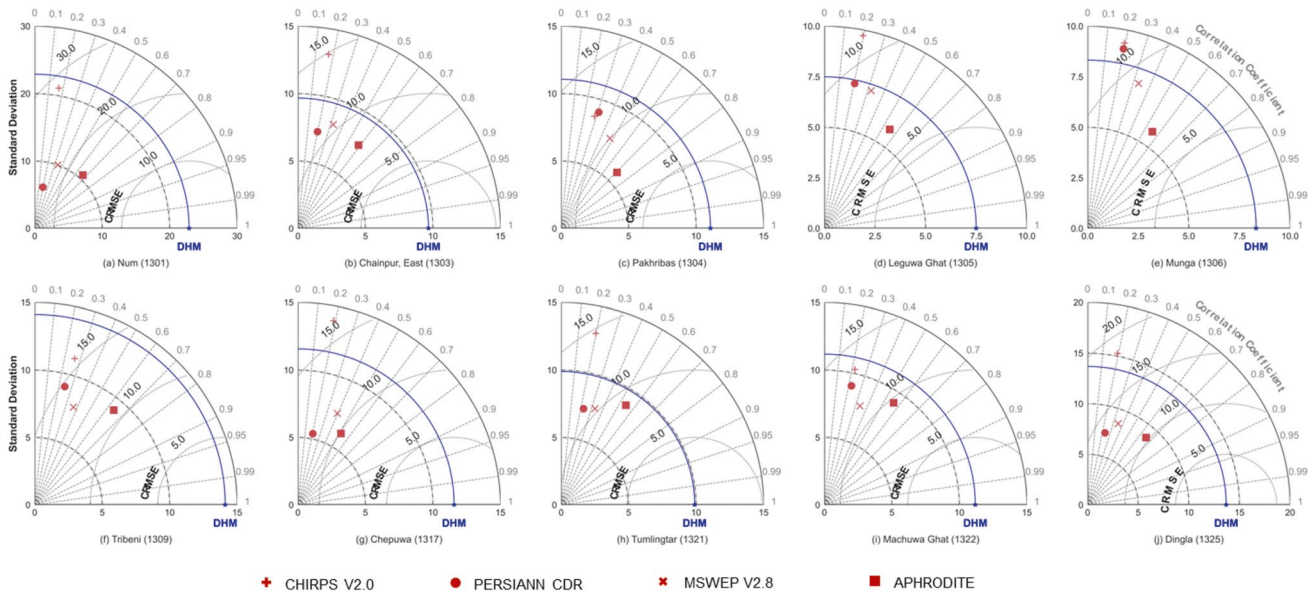


Fig. 4 Taylor diagram of daily precipitation datasets of SPPs versus observed datasets in ARB (1983 to 2014)

Furthermore, in Fig. 4, Taylor diagrams have been plotted to appraise the performance of four SPPs over the observation data at a daily time scale. The Taylor diagram shows that APHRODITE has demonstrated higher correlation and lower CRMSE with all rain gauges than MSWEP V2.8, PERSIANN CDR, and CHIRPS V2.0. APHRODITE, MSWEP V2.8, and PERSIANN CDR have lower SD in all rain gauges except for stations Pakhribas (st. no.: 1304) at elevation 1680 masl and Munga (st. no.: 1306) at elevation 1317 masl. In these two stations, SD, CRMSE, and R^2 values of PERSIANN CDR and CHIRPS V2.0 are nearly identical.

3.1.2 Performance for monthly precipitation

The boxplot of Fig. 5 shows the variation of different continuous statistical indices of monthly precipitation events in ARB (1983 to 2014). The mean monthly precipitation of the observed values of different ground-based stations and the values estimated by various SPPs is shown in Fig. 6.

Table S4 in the supplement section shows various SPPs' performance compared with ground-based stations present on ARB on a monthly scale based on continuous statistical indices. The RMSE values indicate that all four SPPs have monthly biases ranging from a minimum of 35.5 to a maximum of 424.3 mm/month. All of the SPPs have shown high monthly bias in station Num (1497 masl). APHRODITE and CHIRPS V2.0 performed better with low RMSE values compared with PERSIANN CDR and MSWEP V2.8. All of the SPPs have shown a higher correlation with the observed data in the monthly scale comparison, with R^2 values ranging from 0.90 to 0.96 (APHRODITE), 0.80 to 0.89 (MSWEP V2.8), 0.79 to 0.91 (CHIRPS V2.0), and 0.74 to 0.85 (PERSIANN CDR). For all four SPPs, especially in station Num (1497 masl), the ME value ranges from − 287.10 to 63.52, indicating a slightly high error between estimated and observed data. Similarly, the MAE value ranging from 21.8 to 290.3 indicates the presence of error in all four SPPs, among which high error is seen in station Num (1497 masl). Moreover, a mixed pattern of under- and overestimation of the rainfall event can be seen in all SPPs as the RB value ranges from negative to positive. All of the SPPs have highly underestimated the rainfall event in station Num (1497 masl). Performance evaluation performed on the monthly scale shows us APHRODITE and MSWEP V2.8 performing better than CHIRPS V2.0 and PERSIANN CDR in all stations rather than Num and Chepuwa.

Figure 7 shows the Taylor diagram for monthly precipitation. The diagram displays the interrelationship between the estimated and observed dataset's SD, CRMSE, and R^2 employing trigonometric resemblance (Lo Conti et al. 2014). The analysis of the diagram gives us a clear picture that all of the SPPs performed well on the monthly scale,

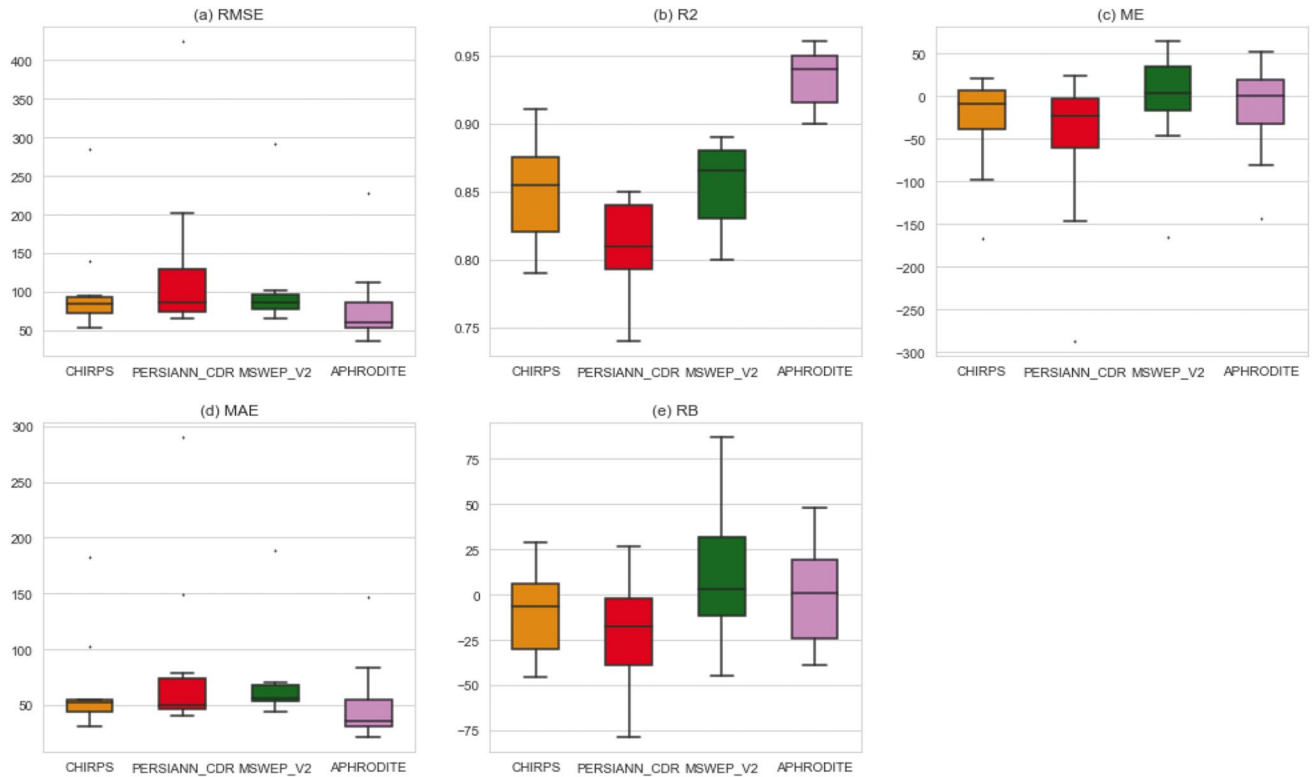


Fig. 5 Continuous statistical indices of monthly precipitation events in ARB (1983 to 2014)

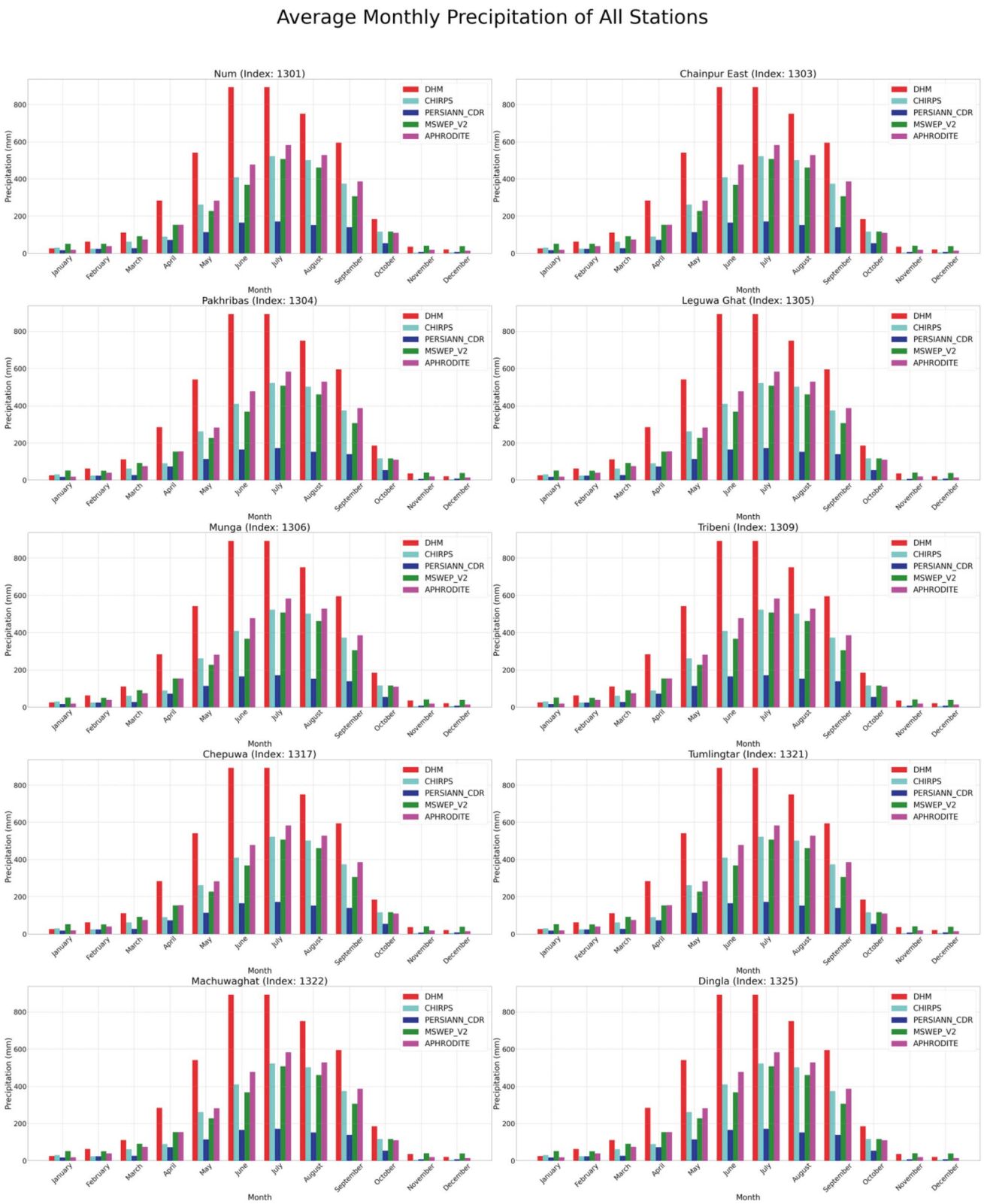


Fig. 6 Mean monthly precipitation based on gauge and SPPs data for the period of 1983–2014 in the ARB

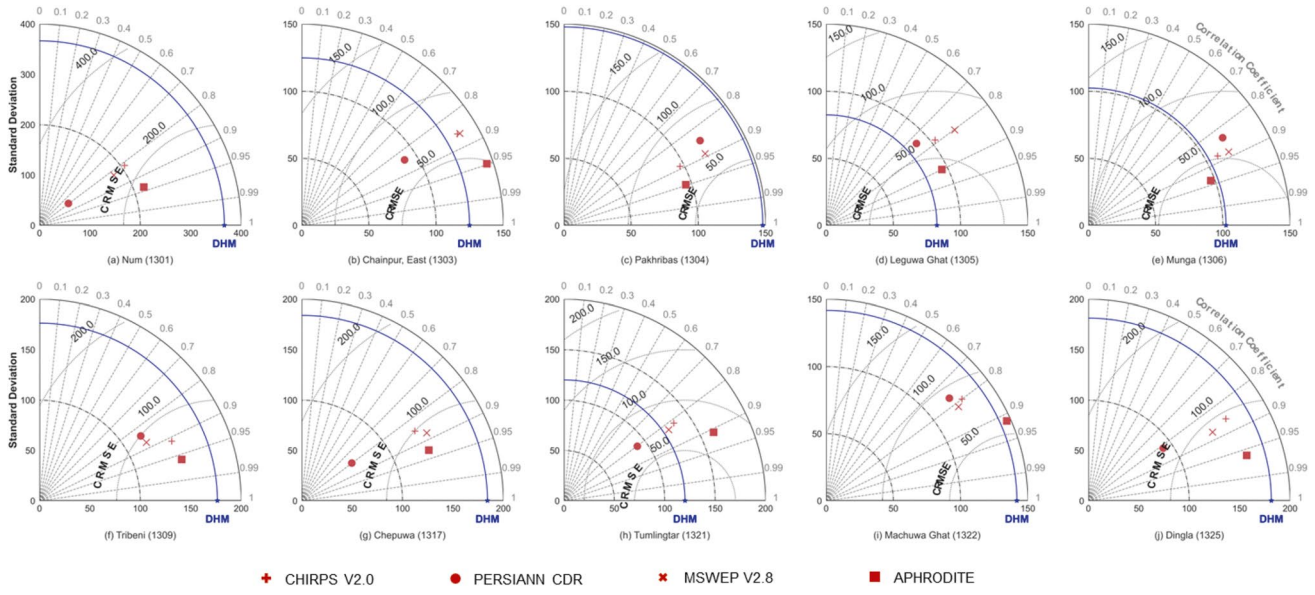


Fig. 7 Taylor diagram of monthly precipitation datasets of SPPs versus observed datasets in ARB (1983 to 2014)

with high correlation values compared to that of the daily scale. In most stations, PERSIANN CDR exhibits the lower SD values except for Pakhribas (st. no.: 1304) at elevation 1680 masl and Munga (st. no.: 1306) at elevation 1317 masl. APHRODITE shows the lower CRMSE values in all stations compared to the other three SPPs.

3.1.3 Performance for annual precipitation

The variation of average annual precipitation of the observed values for all ground-based stations and the values estimated by all four SPPs are shown in Fig. 8.

Table S5 in the supplement section shows the performance of various SPPs in correlation with ground-based stations present in ARB on an annual scale based on continuous statistical indices. The boxplot of Fig. 9 shows the variation of different continuous statistical indices of annual precipitation events in ARB (1983 to 2014). The RMSE values indicate that all four SPPs have high annual biases ranging from a minimum of 158.1 to a maximum of 3500.5 mm/year. Num (1497 masl) and Chepuwa (2590 masl) have higher annual biases than other stations. APHRODITE and MSWEP V2.8 performed slightly better with low RMSE values with respect to CHIRPS V2.0 and PERSIANN CDR. APHRODITE showed the best performance for annual precipitations with a value of R^2 ranging from 0.5 to 0.8, followed by CHIRPS V2.0 ($R^2 = -0.6$ to 0.6), PERSIANN CDR ($R^2 = -0.2$ to 0.6), and MSWEP V2.8 ($R^2 = -0.1$ to 0.5). APHRODITE has shown a higher correlation of 0.8 with the observed data in station Dingla (1190 masl). The ME value for all four SPPs ranges from − 3445.19 to 762.20, indicating a high error between estimated and observed data; especially, the high error is seen in station Num (1497 masl) and Chepuwa (2590 masl). Similarly, the MAE value ranging from 120.80 to 3445.19 indicates an error in all four SPPs, especially the high error seen in station Num (1497 masl) and Chepuwa (2590 masl). A mixed pattern of under- and overestimation of the rainfall event can be seen in all SPPs as the RB value ranges from negative to positive. However, MSWEP V2.8 and APHRODITE were characterized by higher overestimating annual precipitation by 87.1% (Leguwa Ghat, 410 masl) and 48.0% (Tumlingtar, 303 masl), respectively, whereas PERSIANN CDR and CHIRPS V2.0 underestimated the annual precipitation by 78.4% (Num, 1497 masl) and 45.6% (Num, 1497 masl), respectively. Performance evaluation on the annual scale shows that all SPPs performed poorly, especially in Num and Chepuwa.

The Taylor diagram for the annual scale is plotted and shown in Fig. 10. The diagram points out that for station Num (st. no.: 1301) at elevation 1497 masl, the R^2 value of CHIRPS V2.0, PERSIANN CDR, and MSWEP V2.8 is negative, indicating its inverse correlation with the observed data, and for station Chepuwa (st. no: 1317) at elevation 2590 masl, the R^2 value of CHIRPS v2.0 and MSWEP V2.8 is to be negative showing its inverse correlation with the observed data. APHRODITE performed well in all stations with higher R^2 values and lowered SD and CRMSE values. In contrast, in station Machuwa Ghat (st. no: 1322) at elevation 158 masl, CHIRPS V2.0, PERSIANN CDR, and MSWEP V2.8 performed poorly with a very low R^2 value.

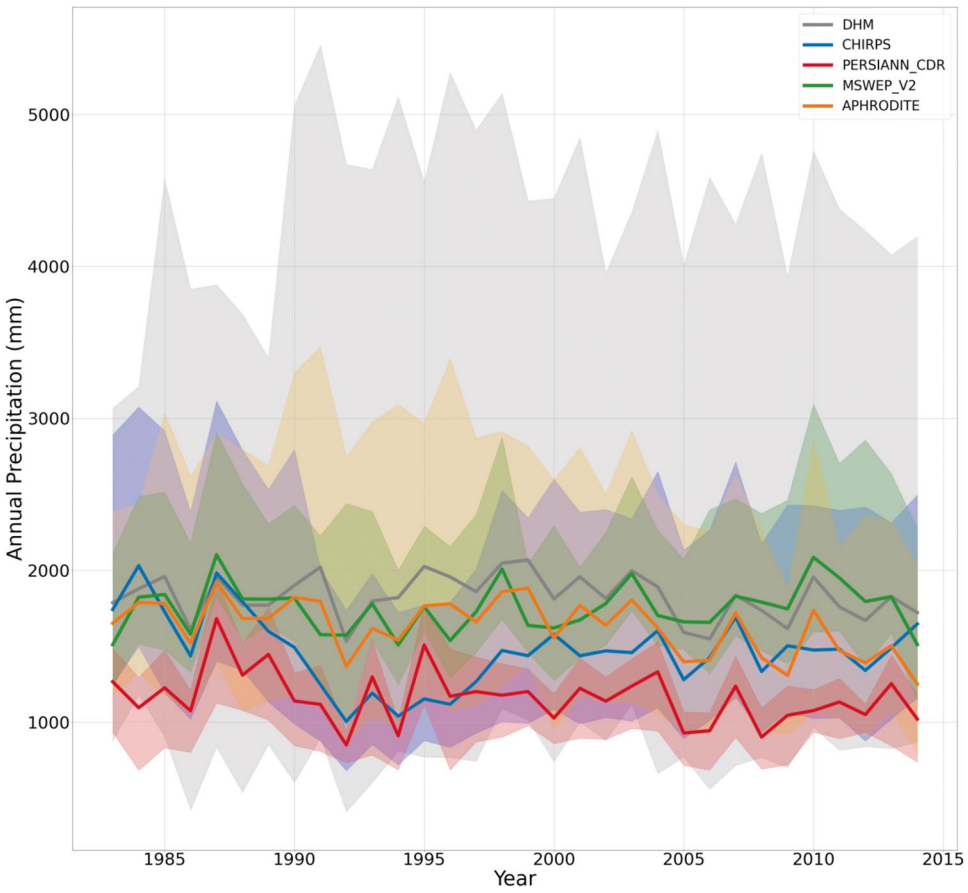


Fig. 8 Variation of average annual precipitation based on all gauges and SPPs data for 1983–2014 in the ARB. The spread represents the range of annual precipitation from the lowest to the highest for all stations distributed across the study area

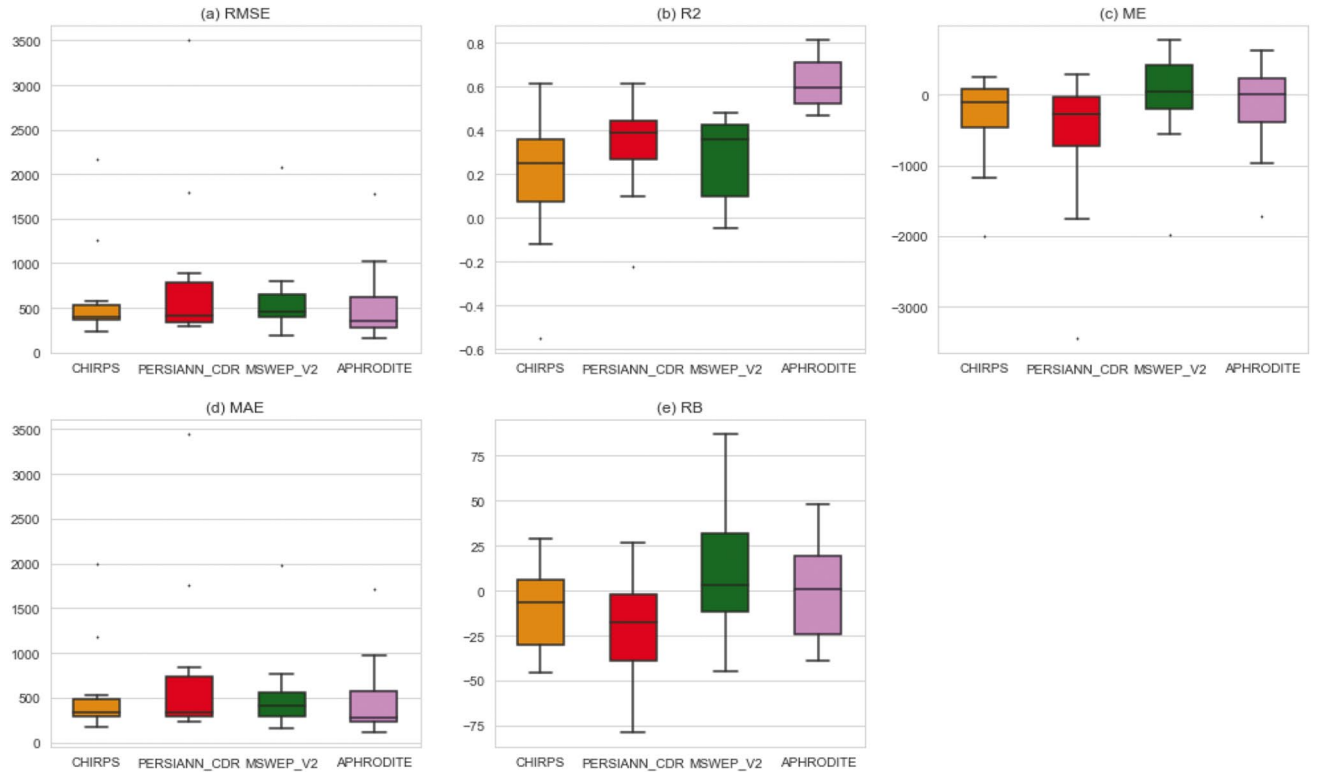


Fig. 9 Continuous statistical indices of annual precipitation events in ARB (1983 to 2014)

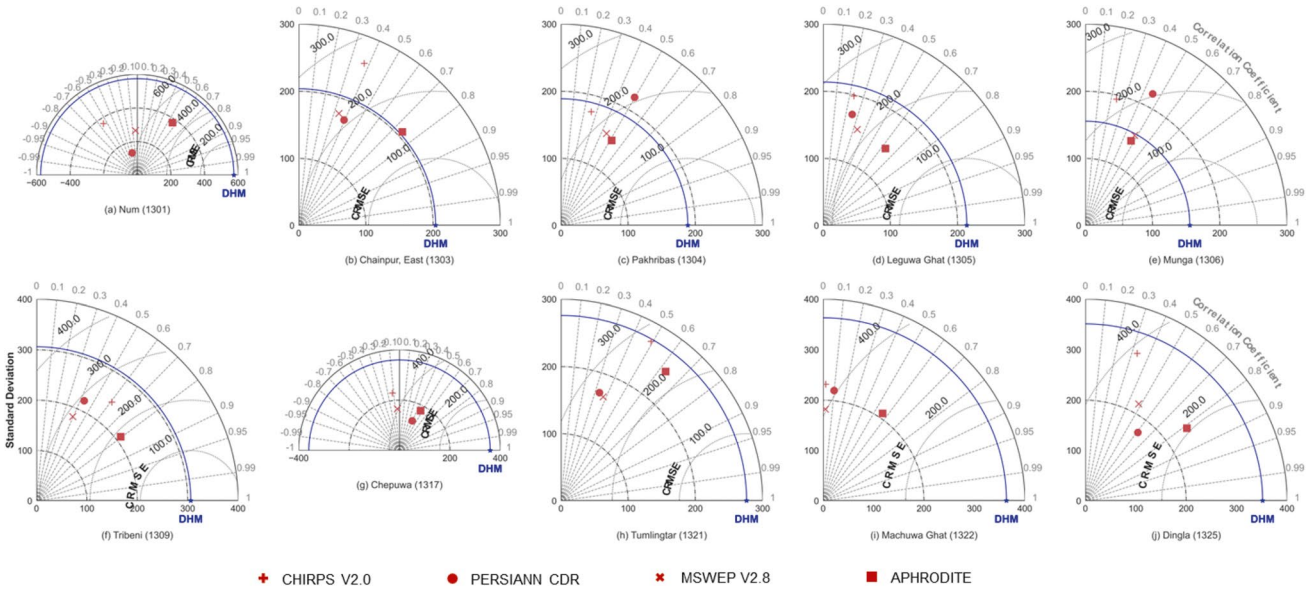


Fig. 10 Taylor diagram of annual precipitation datasets of SPPs versus observed datasets in ARB (1983 to 2014)

3.2 Performance of bias-corrected SSPs

The monthly correction factor for individual stations is determined and applied to correct the estimated values given by four SPPs. The bias correction factors for each station on a monthly scale are presented in Table S6 in the supplement section. The evaluations for bias-corrected values are also done on a daily, monthly, and annual scale.

3.2.1 Performance for daily precipitation

Table S7 in the supplement section shows the evaluation of various SPPs after bias correction at a daily scale compared to precipitation gauges present in the ARB based on continuous statistical indices.

The boxplot of Fig. 11 shows the variation of different continuous statistical indices of daily precipitation events after bias correction in ARB (1983 to 2014). The RMSE values after bias correction for all four SPPs ranged from a minimum of 6.3 to a maximum of 40.8 mm/day, indicating that all four SPPs have daily biases. The performance of APHRODITE has improved with the reduction of RMSE values in all ground stations, whereas CHIRPS V2.0, PERSIANN CDR, and MSWEP V2.8 have shown mixed performance after bias correction where the daily biases of station Num (1497 masl), Pakhribas (1680 masl), Tribeni

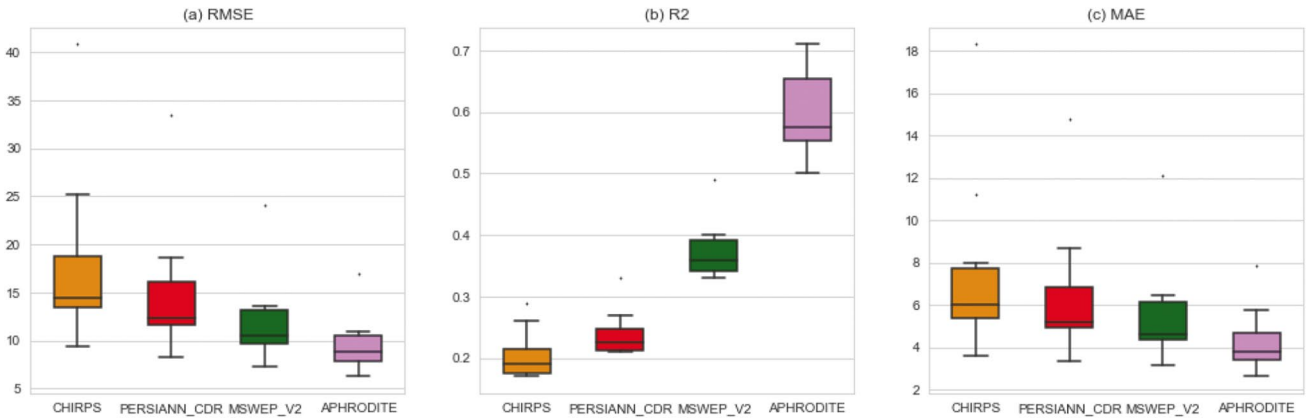


Fig. 11 Continuous statistical indices of daily precipitation events after bias correction in ARB (1983 to 2014). The plot for the ME and RB is not shown in this figure, as their values are zero after bias correction of raw datasets on a daily scale

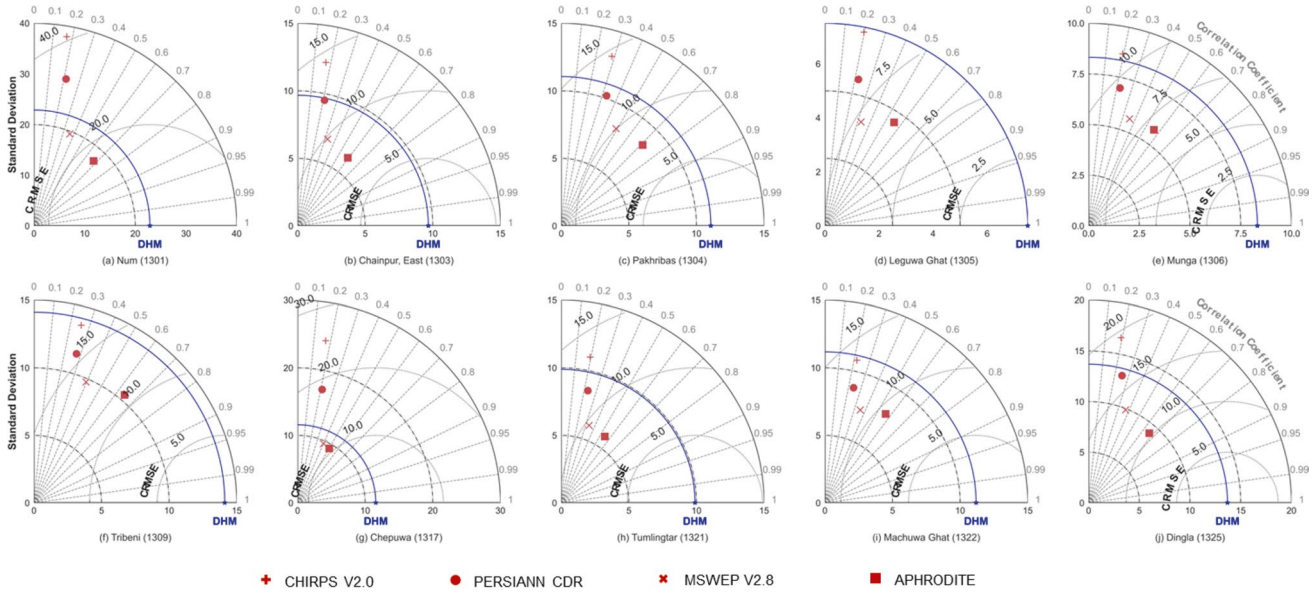


Fig. 12 Taylor diagram of bias-corrected daily precipitation datasets of SPPs versus observed datasets in ARB (1983 to 2014)

(143 masl), Chepuwa (2590 masl), and Dingla (1190 masl) have increased, compared to the raw data of SPPs as shown by RMSE values. Though insignificant, a little improvement (varying from the increment of 0.01 to 0.05) can be seen in R^2 values for all cases. For all four SPPs, the ME value has reduced to zero, indicating consistent results between estimated and observed data.

Similarly, the presence of error in all SPPs has reduced in all stations except for Num (1497 masl), Tribeni (143 masl), Chepuwa (2590 masl), and Machuwaghat (158 masl), as indicated by the MAE values. Moreover, the RB of all SPPs has converged to zero, indicating that rainfall estimation is consistent in all stations. Finally, performance evaluation done after bias correction of all four SPPs indicates that the performance of APHRODITE has improved compared to CHIRPS V2.0, PERSIANN CDR, and MSWEP V2.8.

The Taylor diagram after bias correction of the original datasets of four SPPs has also been generated (Fig. 12). The diagram indicates the improvement of SD, CRMSE, and R^2 for all SPPs, among which APHRODITE has shown satisfactory improvement compared to other SPPs.

3.2.2 Performance for monthly precipitation

The variation of different continuous statistical indices of monthly precipitation events after bias correction in ARB (1983 to 2014) is shown in Fig. 13.

The performance of various SPPs after bias correction compared to ground-based stations present in the ARB on a monthly scale based on continuous statistical indices is presented in Table S8 of the supplement section. The RMSE values have significantly reduced in all stations compared to the raw data, which indicates the reduction of monthly biases present in all four SPPs. Improvement of RMSE values can be seen in all SPPs, among which the RMSE values of APHRODITE and PERSIANN CDR have highly reduced in contrast with CHIRPS V2.0 and MSWEP V2.8. All of the SPPs have shown improved correlation with the observed data in the monthly scale evaluation, with R^2 values ranging from 0.91 to 0.96 (APHRODITE), 0.82 to 0.92 (MSWEP V2.8), 0.81 to 0.92 (CHIRPS V2.0), and 0.79 to 0.89 (PERSIANN CDR). Thus, PERSIANN CDR and MSWEP V2.8 have shown great improvement compared to other SPPs. The ME and RB values for all four SPPs have been reduced to zero. Similarly, the presence of error in all SPPs has reduced distinctively, as indicated by the MAE value. A significant error reduction can be seen in station Num (1497 masl). According to the evaluation done after bias correction, APHRODITE and MSWEP V2.8 have improved performance compared to CHIRPS V2.0 and PERSIANN CDR on the monthly scale.

Figure 14 demonstrates the Taylor diagram plotted for the bias-corrected monthly precipitation datasets of four SPPs versus the observed rainfall datasets. Figure 14 show that all of the SPPs have shown significant improvement after applying the bias correction in all stations of ARB. In addition, all

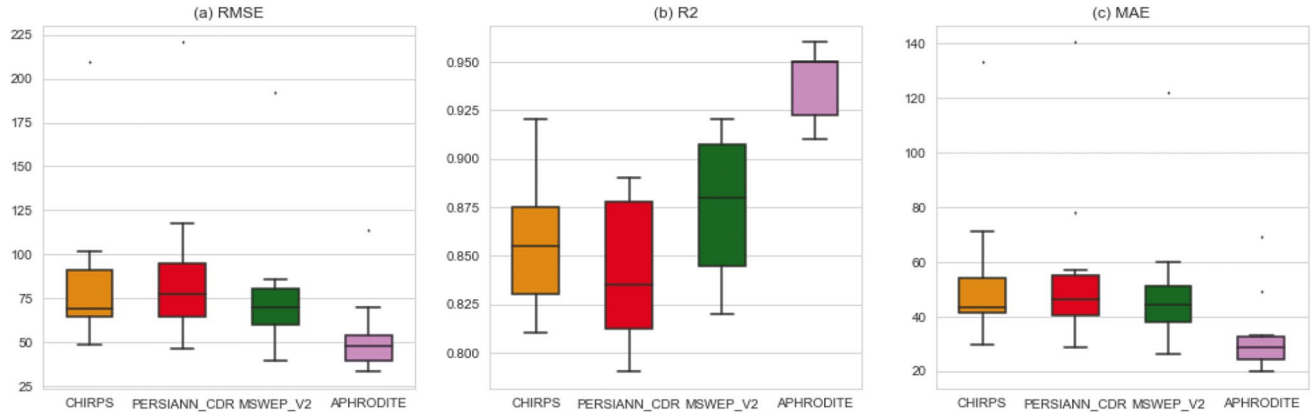


Fig. 13 Continuous statistical indices of monthly precipitation events after bias correction in ARB (1983 to 2014). The plot for the ME and RB is not shown in this figure, as their values are zero after bias correction of raw datasets on a monthly scale

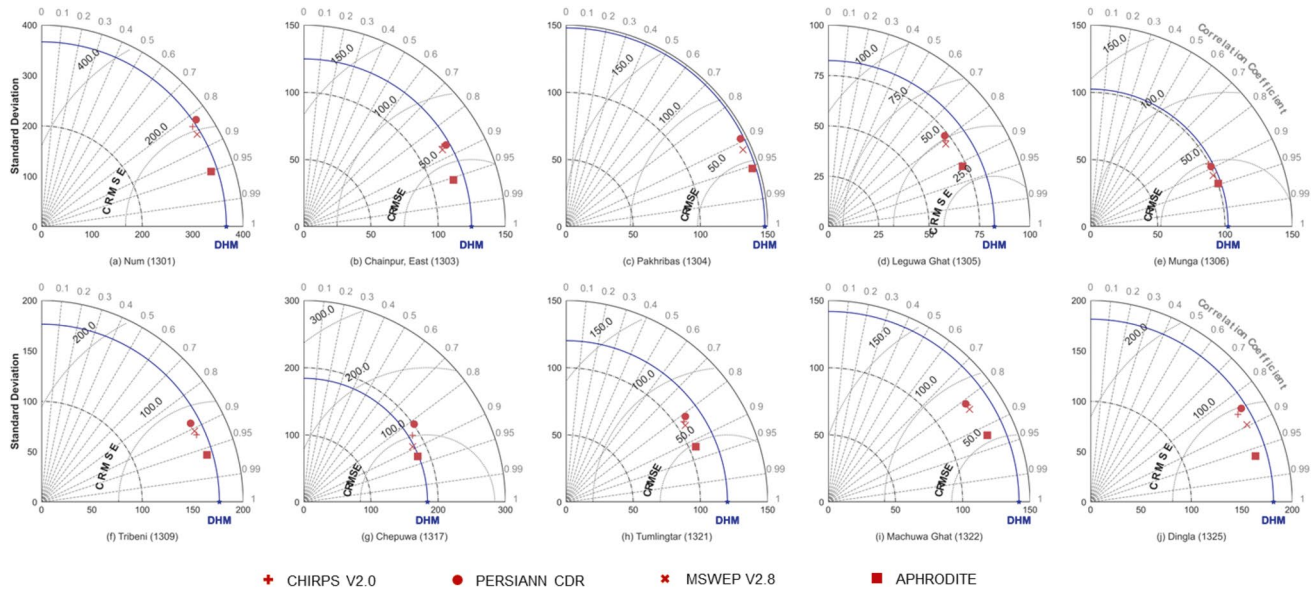


Fig. 14 Taylor diagram of bias-corrected monthly precipitation datasets of SPPs versus observed datasets in ARB (1983 to 2014)

of the SPPs datasets are closer to the gauge observations, as shown by the diagram. The significant improvement in this monthly dataset is due to the correction factor calculated based on the mean monthly scale.

3.2.3 Performance for annual precipitation

The variation of different continuous statistical indices of annual precipitation events after bias correction in ARB (1983 to 2014) is shown in Fig. 15.

Table S9 of the supplement section shows the performance of various SPPs after bias correction with respect to ground-based stations of ARB on an annual scale based on continuous statistical indices. The RMSE values have significantly reduced in all stations compared to the raw data, which indicates the reduction of annual biases present in all four SPPs. Improvement of RMSE values can be seen in all SPPs, among which the RMSE values of PERSIANN CDR, MSWEP V2.8, and APHRODITE have highly reduced in contrast with CHIRPS V2.0, especially in station Num (1497 masl). In addition, all of the SPPs have shown minor improvement in correlation coefficient values. The ME and RB values for all four SPPs have been reduced to zero.

Similarly, the presence of error in all SPPs has reduced distinctively, as indicated by the MAE value. Among all stations, Num (1497 masl) has shown the highest reduction in error. Therefore, according to the evaluation done on the

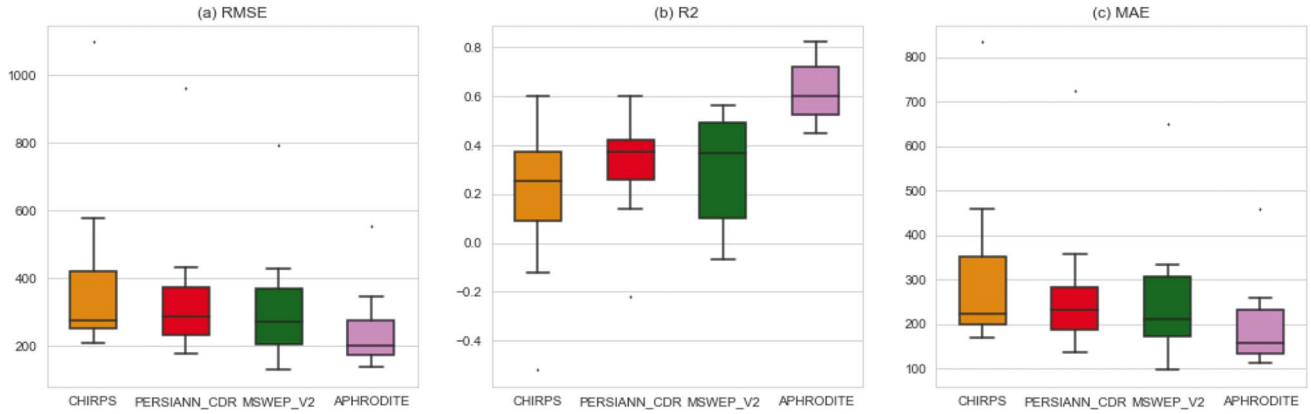


Fig. 15 Continuous statistical indices of annual precipitation events after bias correction in ARB (1983 to 2014). The plot for the ME and RB is not shown in this figure, as their values are zero after bias correction of raw datasets on an annual scale

annual scale, the performance of all SPPs has improved after bias correction. But considering RMSE and MAE values, we can say that PERSIANN CDR, MSWEP V2.8, and APHRODITE performed better than CHIRPS V2.0.

Figure 16 shows a Taylor diagram after the bias correction on an annual scale, based on the SD, CRMSE, and R^2 statistics, for ARB. The diagram indicates that the R^2 values of all SPPs in all stations have not improved considerably. Also, there is a slight improvement in the SD and CRMSE values. APHRODITE performed better after bias correction in all stations compared to other SPPs.

Due to the sparse and uneven distribution of precipitation gauges over unpopulated regions and rugged terrain, SPPs are considered an important source of precipitation data nowadays. Even though SPPs are a good source of precipitation data, they may contain biases that may affect its further application. Therefore, the performance of SPPs must be evaluated for future use of SPPs for hydrological simulation of the basin and for predicting the extreme events that may occur in the basin. For example, Num (1497 masl) receives a high amount of rainfall with an average annual rainfall of 4392.69 mm/year among all other ground-based stations. However, all SPPs underestimated the rainfall event in station Num (1301), which improved after applying bias correction. The evaluation of SPPs on a daily scale shows that APHRODITE performed better, followed by MSWEP V2.8, PERSIANN CDR, and CHIRPS V2.0. On a monthly scale, APHRODITE performed better, followed by MSWEP V2.8, CHIRPS V2.0, and PERSIANN CDR. APHRODITE performed better in the annual scale evaluation, followed by

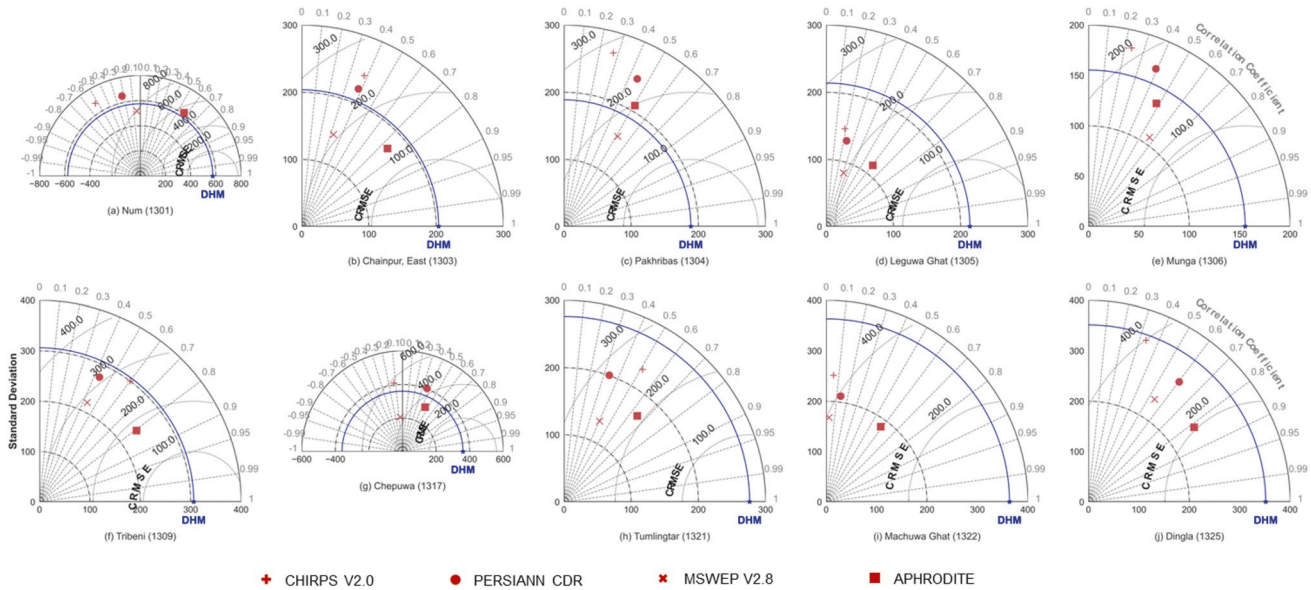


Fig. 16 Taylor diagram of bias-corrected annual precipitation datasets of SPPs versus observed datasets in ARB (1983 to 2014)

PERSIANN CDR, CHIRPS V2.0, and MSWEP V2.8. Bias correction done for these SPPs has significantly improved the performance of APHRODITE and MSWEP V2.8 compared to CHIRPS V2.0 and PERSIANN CDR.

The main reason for APHRODITE performing better in all time scales is that APHRODITE's long-term daily gridded precipitation datasets have been developed from ground-based rain gauge datasets over Asia in collaboration with local meteorological agencies and researchers (Yatagai et al. 2012). In addition, in APHRODITE V1101, the developers had made huge efforts in collecting more dense data, and the methods for analysis and quality control have also been improved (Yatagai et al. 2012). The precipitation records used in APHRODITE are mainly obtained from the Global Telecommunication System (GTS), individual collections, and existing meteorological station record datasets, such as the Global Historical Climatology Network (GHCN) and the National Climatic Data Center (NCDC). APHRODITE seems to be a promising interpolated data that use local precipitation information. However, the quality of the gridded data is highly dependent on the amount of input data, which varies in space and time. Therefore, APHRODITE and MSWEP V2.8 are highly recommended for further hydrological simulations and extreme events prediction for the selected study area.

4 Conclusions

We have conducted a performance evaluation of three SPPs and one observation-based gridded precipitation product over the Arun River Basin. This study employed the bias correction of those SPPs using the linear scaling method. The methods applied are scalable to other regions across the globe. Based on the current study results, we conclude as follows:

i) APHRODITE and PERSIANN CDR have shown high accuracy of rainfall events with respect to observed datasets, and all of the SPPs have shown significant false alarms. CHIRPS V2.0 underestimated the number of precipitation events for the entire study region; PERSIANN CDR showed the mixed under- and overestimation pattern, whereas MSWEP V2.8 and APHRODITE overestimated the number of precipitation events for the entire study area. All four SPPs have daily, monthly, and annual positive biases. APHRODITE shows a higher correlation with the observed data on a daily and annual scale. In contrast, all SPPs have shown a higher correlation with the observed data on the monthly scale. The SPPs exhibit the error between estimated and observed data. Also, all chosen SPPs have generated mixed (i.e., under- and overestimation) patterns of the rainfall events. We, therefore, suggest that the use of those SPPs in their raw form might have uncertainty, and the biases need to be reduced using appropriate bias correction techniques.

ii) Bias correction by linear scaling method has significantly improved the performance of all SPPs compared to the data before correction. For example, bias corrections have significantly improved the performance of APHRODITE and MSWEP V2.8 compared to CHIRPS V2.0 and PERSIANN CDR. Hence, APHRODITE and MSWEP V2.8 are recommended for further implementation for hydrological simulations and extreme events prediction in the ARB.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00704-022-04001-y.

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Author contribution SD, RT, and VPP visualized the research work; SD formulated the methodology and analyzed the data; RT helped in coding done in Python; RT and VPP supervised the research work; SD wrote the original draft report; and RT and VPP helped in reviewing and editing. All authors have reviewed and approved the manuscript.

Availability of data and material The collection of daily rainfall datasets at the selected precipitation stations was done from the Department of Hydrology and Meteorology (DHM), Nepal (http://www.dhm.gov.np). The data is not available publicly. These data have to be purchased (http://dhm.gov.np/pricelist.html) and used by the person(s)/institution for the sole purpose of their work as authorized by the DHM. The data cannot be used for commercial purposes. Sources of chosen satellite-based precipitation products are mentioned in the text.

Code availability This study does not employ any customized code. All tools used in this study are mentioned in the Data and Methodology section.

Declarations

Ethics approval The authors are responsible for the manuscript's integrity, including scientific ethics.

Consent to participate There are no participants other than authors in this study. The authors agree to participate.

Consent for publication There are no participants other than authors in this study for their consent on publication. The authors agree to publish.

Conflict of interest The authors declare no competing interests.

References

Abd Elhamid AMI, Eltahan AMH, Mohamed LME, Hamouda IA (2020) Assessment of the two satellite-based precipitation products TRMM and RFE rainfall records using ground based measurements. Alex Eng J 59(2):1049–1058. https://doi.org/10.1016/j.aej.2020.03.035

Amorim J da S, Viola MR, Junqueira R, de Oliveira VA, de Mello CR (2020) Evaluation of satellite precipitation products for hydrological modeling in the Brazilian cerrado biome. Water (Switzerland), 12(9). https://doi.org/10.3390/W12092571

Ashouri H, Hsu KL, Sorooshian S, Braithwaite DK, Knapp KR, Cecil LD, Nelson BR, Prat OP (2015) PERSIANN-CDR: daily precipitation climate data record from multisatellite observations for hydrological and climate studies. Bull Am Meteor Soc 96(1):69–83. https://doi.org/10.1175/BAMS-D-13-00068.1

Athey A, Ellis AW, Carroll DF, Blevins RD (2015) Verification of satellite derived precipitation estimates over complex terrain: a ground truth analysis for Nepal. 78. https://vtechworks.lib.vt.edu/bitstream/handle/10919/52917/Athey_AT_T_2015.pdf;sequence=1. Accessed 15 Sept 2021

Bajracharya SR, Shrestha MS, Shrestha AB (2017) Assessment of high-resolution satellite rainfall estimation products in a streamflow model for flood prediction in the Bagmati Basin, Nepal. J Flood Risk Manag 10(1):5–16. https://doi.org/10.1111/jfr3.12133

Bajracharya SR, Palash W, Shrestha MS, Khadgi VR, Duo C, Das PJ, Dorji C (2015) Systematic evaluation of satellite-based rainfall products over the Brahmaputra basin for hydrological applications. Adv Meteorol. https://doi.org/10.1155/2015/398687

Beck HE, Wood EF, Pan M, Fisher CK, Miralles DG, Van Dijk AIJM, McVicar TR, Adler RF (2019) MSWep v2 global 3-hourly 01° precipitation: methodology and quantitative assessment. Bull Amer Meteorol Soc 100(3):473–500. https://doi.org/10.1175/BAMS-D-17-0138.1

Belayneh A, Sintayehu G, Gedam K, Muluken T (2020) Evaluation of satellite precipitation products using HEC-HMS model. Model Earth Syst Environ 6(4):2015–2032. https://doi.org/10.1007/s40808-020-00792-z

Bookhagen B, Burbank DW (2010) Toward a complete Himalayan hydrological budget: spatiotemporal distribution of snowmelt and rainfall and their impact on river discharge. J Geophys Res Earth Surf 115(3):1–25. https://doi.org/10.1029/2009JF001426

Chen J, Brissette FP, Chaumont D, Braun M (2013) Finding appropriate bias correction methods in downscaling precipitation for hydrologic impact studies over North America. Water Resour Res 49(7):4187–4205. https://doi.org/10.1002/wrcr.20331

Chen J, Li Z, Li L, Wang J, Qi W, Xu CY, Kim JS (2020) Evaluation of multi-satellite precipitation datasets and their error propagation in hydrological modeling in a monsoon-prone region. Remote Sens 12(21):1–33. https://doi.org/10.3390/rs12213550

Dinku T, Ruiz F, Connor SJ, Ceccato P (2010) Validation and intercomparison of satellite rainfall estimates over Colombia. J Appl Meteorol Climatol 49(5):1004–1014. https://doi.org/10.1175/2009JAMC2260.1

Duncan JMA, Biggs EM (2012) Assessing the accuracy and applied use of satellite-derived precipitation estimates over Nepal. Appl Geogr 34:626–638. https://doi.org/10.1016/j.apgeog.2012.04.001

Funk C, Peterson P, Landsfeld M, Pedreros D, Verdin J, Shukla S, Husak G, Rowland J, Harrison L, Hoell A, Michaelsen J (2015) The climate hazards infrared precipitation with stations - a new environmental record for monitoring extremes. Scientific Data 2:1–21. https://doi.org/10.1038/sdata.2015.66

Ghaemi E, Foelsche U, Kann A, Fuchsberger J (2021) Evaluation of INCA precipitation analysis using a very dense rain gauge network in southeast Austria. Hydrology and Earth System Sciences Discussions, February, 1–33. 10.5194/hess-2021-34

Ghaju S, Alfredsen K (2016) Evaluation of satellite based precipitations and their applicability for rainfall runoff modelling in Narayani basin of Nepal. J Hydrol Meteorol 8(1):22–31. https://doi.org/10.3126/jhm.v8i1.15569

Hamal K, Sharma S, Khadka N, Baniya B, Ali M, Shrestha MS, Xu T, Shrestha D, Dawadi B (2020) Evaluation of MERRA-2 precipitation products using gauge observation in Nepal. Hydrology 7(3):1–21. https://doi.org/10.3390/hydrology7030040

Hossain K, Water S, Agency S (2016) Satellite based flood forecasting for the Koshi River Basin , Nepal TITLE PAGE Satellite based flood forecasting for the Koshi River Basin , Nepal. April 2014. https://doi.org/10.13140/RG.2.1.2112.6807

Hsu KL, Gao X, Sorooshian S, Gupta HV (1997) Precipitation estimation from remotely sensed information using artificial neural networks. J Appl Meteorol 36(9):1176–1190. https://doi.org/10.1175/1520-0450(1997)036%3c1176:PEFRSI%3e2.0.CO;2

Islam MN, Das S, Uyeda H (2010) Calibration of TRMM derived rainfall over Nepal during 1998–2007. In The Open Atmospheric Science Journal (Vol. 4). http://trmm.gsfc.nasa.gov/3b42.html. Accessed 15 Dec 2021

Ji X, Li Y, Luo X, He D, Guo R, Wang J, Bai Y, Yue C, Liu C (2020) Evaluation of bias correction methods for APHRODITE data to improve hydrologic simulation in a large Himalayan basin. Atmos Res 242(September 2019):104964. https://doi.org/10.1016/j.atmosres.2020.104964

Jiang S, Liu S, Ren L, Yong B, Zhang L, Wang M, Lu Y, He Y (2017) Hydrologic evaluation of six high resolution satellite precipitation products in capturing extreme precipitation and streamflow over a medium-sized basin in China. Water (Switzerland), 10(1), 1–17. https://doi.org/10.3390/w10010025

JICA (1985) Master plan study on Kosi River water resources development—interim report, Tokyo

Karki R, Talchabhadel R, Aalto J, Baidya SK (2015) New climatic classification of Nepal. Theor Appl Climatol 125(3):799–808. https://doi.org/10.1007/S00704-015-1549-0

Kattelmann R (1990) Hydrology and development of the Arun River, Nepal. Hydrol Mt Reg I 193:777–784

Khatakho R, Talchabhadel R, Thapa BR (2021) Evaluation of different precipitation inputs on streamflow simulation in Himalayan River basin. J Hydrol 599(October 2020):126390. https://doi.org/10.1016/j.jhydrol.2021.126390

Kidd C, Becker A, Huffman GJ, Muller CL, Joe P, Skofronick-Jackson G, Kirschbaum DB (2017) So, how much of the Earth's surface is covered by rain gauges? Bull Am Meteor Soc 98(1):69–78. https://doi.org/10.1175/BAMS-D-14-00283.1

Krakauer NY, Pradhanang SM, Lakhankar T, Jha AK (2013) Evaluating satellite products for precipitation estimation in mountain regions: a case study for Nepal. Remote Sens 5(8):4107–4123. https://doi.org/10.3390/rs5084107

Lo Conti F, Hsu KL, Noto LV, Sorooshian S (2014) Evaluation and comparison of satellite precipitation estimates with reference to a local area in the Mediterranean Sea. Atmos Res 138:189–204. https://doi.org/10.1016/j.atmosres.2013.11.011

Mashingia F, Mtalo F, Bruen M (2014) Validation of remotely sensed rainfall over major climatic regions in northeast Tanzania. Phys Chem Earth 67–69(January):55–63. https://doi.org/10.1016/j.pce.2013.09.013

Masood M, Shakir AS, Azhar AH, Nabi G, Habib-u-Rehman (2020) Assessment of real time, multi-satellite precipitation products under diverse climatic and topographic conditions. Asia-Pacific J Atmos Sci, 56(4), 577–591. https://doi.org/10.1007/s13143-019-00166-1

Mazzoleni M, Brandimarte L, Amaranto A (2019) Evaluating precipitation datasets for large-scale distributed hydrological modelling. J Hydrol 578(December 2018):24076. https://doi.org/10.1016/j.jhydrol.2019.124076

Miao C, Ashouri H, Hsu KL, Sorooshian S, Duan Q (2015) Evaluation of the PERSIANN-CDR daily rainfall estimates in capturing the behavior of extreme precipitation events over China. J Hydrometeorol 16(3):1387–1396. https://doi.org/10.1175/JHM-D-14-0174.1

Moazami S, Golian S, Kavianpour MR, Hong Y (2014) Uncertainty analysis of bias from satellite rainfall estimates using copula method. Atmos Res 137:145–166. https://doi.org/10.1016/j.atmosres.2013.08.016

Müller MF, Thompson SE (2013) Bias adjustment of satellite rainfall data through stochastic modeling: methods development and application to Nepal. Adv Water Resour 60:121–134. https://doi.org/10.1016/j.advwatres.2013.08.004

Nepal B, Shrestha D, Sharma S, Shrestha MS, Aryal D, Shrestha N (2021) Assessment of GPM-era satellite products' (IMERG and GSMaP) ability to detect precipitation extremes over mountainous country Nepal. Atmosphere 12(2):254. https://doi.org/10.3390/atmos12020254

Nguyen P, Ombadi M, Sorooshian S, Hsu K, AghaKouchak A, Braithwaite D, Ashouri H, Rose Thorstensen A (2018) The PERSIANN family of global satellite precipitation data: a review and evaluation of products. Hydrol Earth Syst Sci 22(11):5801–5816. https://doi.org/10.5194/hess-22-5801-2018

Olen SM, Bookhagen B, Hoffmann B, Sachse D, Adhikari DP, Strecker MR, Olen SM, Bookhagen B, Hoffmann B, Sachse D, Adhikari DP, M. R. S. (2014) Understanding erosion rates in the Himalayan orogen: a case study from the Arun Valley. J Geophys Res F Earth Surf, 120(10), 300–316. https://doi.org/10.1002/2013JF002871. Received

Sharifi E, Eitzinger J, Dorigo W (2019) Performance of the state-of-the-art gridded precipitation products over mountainous terrain : a regional study over Austria. 1–20.

Sharma S, Chen Y, Zhou X, Yang K, Li X, Niu X, Hu X, Khadka N (2020a) Evaluation of GPM-era satellite precipitation products on the southern slopes of the central Himalayas against rain gauge data. Remote Sens 12(11). https://doi.org/10.3390/rs12111836

Sharma S, Khadka N, Hamal K, Shrestha D, Talchabhadel R, Chen Y (2020b) How accurately can satellite products (TMPA and IMERG) detect precipitation patterns, extremities, and drought across the Nepalese Himalaya? Earth Space Sci 7(8). https://doi.org/10.1029/2020bEA001315

Shrestha MS, Artan GA, Bajracharya SR, Gautam DK, Tokar SA (2011) Bias-adjusted satellite-based rainfall estimates for predicting floods: Narayani Basin. J Flood Risk Manag 4(4):360–373. https://doi.org/10.1111/j.1753-318X.2011.01121.x

Shrestha MS, Artan GA, Bajracharya SR, Sharma RR (2008) Using satellite-based rainfall estimates for streamflow modelling: Bagmati Basin. J Flood Risk Manag 1(2):89–99. https://doi.org/10.1111/j.1753-318x.2008.00011.x

Shrestha M, Takara K, Kubota T, Bajracharya S (2011) Verification of GSMaP rainfall estimates over the Central Himalayas. J Japan Soc Civ Eng Ser B1 67(4):I_37-I_42. https://doi.org/10.2208/jscejhe.67.i_37

Shrestha M, Acharya SC, Shrestha PK (2017a) Bias correction of climate models for hydrological modelling–are simple methods still useful? Meteorol Appl 24(3):531–539. https://doi.org/10.1002/met.1655

Shrestha NK, Qamer FM, Pedreros D, Murthy MSR, Wahid SM, Shrestha M (2017b) Evaluating the accuracy of Climate Hazard Group (CHG) satellite rainfall estimates for precipitation based drought monitoring in Koshi basin, Nepal. J Hydrol Reg Stud 13(February):138–151. https://doi.org/10.1016/j.ejrh.2017.08.004

Shrestha TB (1989) Development of ecology of the Arun River Basin in Nepal

Sun R, Yuan H, Liu X, Jiang X (2016) Evaluation of the latest satellite-gauge precipitation products and their hydrologic applications over the Huaihe River basin. J Hydrol 536:302–319. https://doi.org/10.1016/j.jhydrol.2016.02.054

Talchabhadel R, Aryal A, Kawaike K, Yamanoi K, Nakagawa H, Bhatta B, Karki S, Thapa BR (2021) Evaluation of precipitation elasticity using precipitation data from ground and satellite-based estimates and watershed modeling in western Nepal. J Hydrol Reg Stud, 33(July 2020), 100768. https://doi.org/10.1016/j.ejrh.2020.100768

Talchabhadel R, Karki R, Parajuli B (2017) Intercomparison of precipitation measured between automatic and manual precipitation gauge in Nepal. Measurement 106:264–273. https://doi.org/10.1016/J.MEASUREMENT.2016.06.047

Tan ML, Ibrahim AL, Duan Z, Cracknell AP, Chaplot V (2015) Evaluation of six high-resolution satellite and ground-based precipitation products over Malaysia. Remote Sens 7(2):1504–1528. https://doi.org/10.3390/rs70201504

Tan ML, Santo H (2018) Comparison of GPM IMERG, TMPA 3B42 and PERSIANN-CDR satellite precipitation products over Malaysia. Atmos Res 202(July2017):63–76. https://doi.org/10.1016/j.atmosres.2017.11.006

Taylor KE (2001) in a Single Diagram. 106, 7183–7192

Tian Y, Peters-Lidard CD, Eylander JB (2010) Real-time bias reduction for satellite-based precipitation estimates. J Hydrometeorol 11(6):1275–1285. https://doi.org/10.1175/2010JHM1246.1

Tong, K., Su, F., Yang, D., & Hao, Z. (2014). Evaluation ofsatellite precipitation retrievals and their potential utilities in hydrologic modeling over the Tibetan Plateau. J Hydrol, 519(PA), 423–437. https://doi.org/10.1016/j.jhydrol.2014.07.044

Venkatesh K, Krakauer NY, Sharifi E, Ramesh H (2020) Evaluating the performance of secondary precipitation products through statistical and hydrological modeling in a mountainous tropical basin of India. Adv Meteorol. https://doi.org/10.1155/2020/8859185

Wang Q, Xia J, She D, Zhang X, Liu J, Zhang Y (2021) Assessment of four latest long-term satellite-based precipitation products in capturing the extreme precipitation and streamflow across a humid region of southern China. Atmos Res 257(8):105554. https://doi.org/10.1016/j.atmosres.2021.105554

WMO (2008) Guide to hydrological practices volume I hydrology – from measurement to hydrological information. In WMO-No. 168 (Vol. 53, Issue 9).

Yatagai A, Kamiguchi K, Arakawa O, Hamada A, Yasutomi N, Kitoh A (2012) Aphrodite constructing a long-term daily gridded precipitation dataset for Asia based on a dense network of rain gauges. Bull Am Meteor Soc 93(9):1401–1415. https://doi.org/10.1175/BAMS-D-11-00122.1

Zhang Y, Hanati G, Danierhan S, Liu Q, Xu Z (2020) Evaluation and comparison of daily gpm/trmm precipitation products over the Tianshan mountains in China. Water (Switzerland), 12(11). https://doi.org/10.3390/w12113088

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